Two motor-challenge items as a candidate cross-context suggestibility screener

W. Weaver

Word count: ≈4,200 Figures: 5 Tables: 4

Secondary reanalysis of publicly available data. Not preregistered. No conflicts of interest. No funding.

Abstract

Full suggestibility assessment is often impractical in screening contexts. We asked whether two motor-challenge items—arm rigidity and arm immobilisation—show enough cross-context validity to motivate a standalone first-pass screener. Applying a consistent analysis to five open datasets (N = 240–584) spanning non-hypnotic (PCS), hypnotic (SWASH), and classic group-administered (HGSHS:A) contexts, the motor composite correlated r = 0.78–0.85 with the full-scale total (intensity and involuntariness scales; r = 0.68 on the objective binary) and r = 0.55–0.69 with the remaining non-motor items (overlap-free). Music hallucination and negative visual hallucination approach floor in unselected samples (80–93% at zero); with near-universal zero scores they add little to between-person variance on the total, which is therefore disproportionately shaped by higher-variance challenge items. Classification weighted κ ranged 0.50–0.55 (PCS) and 0.65 (SWASH and HGSHS:A involuntariness) against the full total, and 0.29–0.52 against the overlap-free rest score. Against external criteria with no item overlap, the motor pair retained 71–80% of the full scale's predictive signal. The pattern holds across contexts and measurement axes. These results support prospective validation of the motor pair as a standalone first-pass screener in unselected samples.

Keywords: hypnotic suggestibility; phenomenological control; SWASH; PCS; HGSHS:A; short form; screening; floor effects

1 Introduction

Full administration of standard suggestibility scales—PCS (Lush et al., 2021), SWASH (Lush et al., 2018), HGSHS:A (Shor and Orne, 1962)—requires ten or more suggestion items, structured administration, and per-item ratings. For screening, online intake, or time-limited contexts, this is often not feasible.

Whether such administration is necessary depends on where between-person variance actually concentrates. A long-standing critique observes that item difficulty is uneven: several suggestions are passed by very few people in unselected samples and therefore carry little discriminating information (Coe and Sarbin, 1971; Sadler and Woody, 2004). Bifactor analysis of HGSHS:A supports a general factor structure with minor domain-specific components (Zahedi and Sommer, 2022), though whether a unitary general factor underlies suggestibility scales more broadly remains an open question. Principal component analysis of the normative PCS and SWASH data shows a consistent first-component structure within those scales (Lush et al., 2021, supplementary materials): the motor pair (arm rigidity, arm immobilisation) loads most strongly on the first component in both conditions (PCS: 0.706–0.727; SWASH: 0.782–0.803), while music hallucination and negative visual hallucination load weakly (0.19–0.41 in PCS), consistent with their near-floor rates. On the second component, passive ideomotor items (hand lowering, hands together) and perceptual items (mosquito, visual hallucination) occupy opposite poles; the motor challenge items sit between them, near-neutral on this dimension—especially under induction, where arm rigidity loads -0.05 on PC2. The motor pair is structurally central: highest on the first component, minimally pulled toward either secondary pole, and consistent with indexing a broad shared factor if one exists.

If variance concentrates in the motor pair, a two-item composite may serve as a first-pass screener. A motor-focused five-item short form of HGSHS:A achieves $r = 0.83$ and $\kappa \approx 0.58$ against its parent (Riegel et al., 2021; Zech et al., 2024). What has not been examined is whether a similar structure holds across hypnotic and non-hypnotic contexts, across intensity vs. involuntariness measurement axes, and whether a two-item composite produces defensible classification metrics.

We apply a consistent pipeline to five open datasets spanning all three contexts. We report part-whole and overlap-free correlations, classification metrics, an all-pairs comparison, external validity tests, and retest stability. The motor pair items are arm rigidity and arm immobilisation: mid-scale in unselected samples, high variance, behaviourally observable.

2 Method

2.1 Datasets

Five open datasets were analysed (Table 1). PCS and SWASH use identical ten-item suggestion sets rated 0–5 for subjective intensity; they differ only in administration context. HGSHS:A uses the Bowers Involuntariness Scale (Bowers, 1981) (0 = did not experience; 1 = voluntary; 5 = involuntary-automatic) and objective pass/fail; it is analysed separately and never pooled with the intensity datasets. Dataset-provided exclusion variables were applied, removing $n = 6$ participants from the Norms sample, $n = 18$ from the HGSHS:A sample, and $n = 62$ from the vEAR-PCS sample. The SWASH validation sample was additionally restricted to participants with non-missing first-test scores ($n = 77$ excluded).

Table 1: Dataset summary.

Sample	Scale	Context	Dimension	N_{raw}	N_{analytic}
Norms-PCS	PCS	Non-hypnotic	Intensity	490	244
Norms-SWASH	SWASH	Hypnotic	Intensity	490	240
PCS-VVIQ	PCS	Non-hypnotic	Intensity	508	508
SWASH-val	SWASH	Hypnotic	Intensity	495	418
HGSHS:A	HGSHS:A	Classic group	Inv.+Obj.	602	584
vEAR-PCS	PCS	Non-hypnotic	Intensity	514	452

Note. Norms sample: participants randomly assigned to PCS or SWASH condition; each participant contributed to one condition. OSF accession codes in Open Science Statement.

2.2 Scoring

Author-precomputed totals were used for the Norms, SWASH validation, and vEAR-PCS samples. For the PCS-VVIQ sample, the canonical 10-item mean was computed with geometric-mean post-hypnotic suggestion ($\sqrt{\text{urge} \times \text{amnesia}}$) and arithmetic-mean taste.

The motor composite is the sum of the arm rigidity and arm immobilisation items. Source column names: Norms sample: `ArmRigiditySubjectiveRating` + `ArmImmobilisationSubjectiveRating`; PCS-VVIQ and vEAR-PCS samples: `StiffArm` + `HandAndArmHeavy`; SWASH validation sample: `Q5RIGIDITYSUBJ` + `Q6IMMOBILESUBJ`; HGSHS:A sample: `INV.6` + `INV.4` (involuntariness) or `HGSHS.A6` + `HGSHS.A4` (objective).

2.3 Analyses

Two correlation metrics are reported per dataset: r_{full} (motor composite vs. full-scale total; motor items inside the total) and r_{rest} (motor composite vs. the remaining non-motor items, no shared items). Both are always reported together. Sum and mean scoring are linear transforms and produce identical correlations. Itemwise z-scored composites were checked as a sensitivity analysis and produced materially unchanged results. Listwise deletion changes nothing. Spearman-Brown reliability $= 2r/(1 + r)$, where r is the inter-item correlation (Eisinga et al., 2013). All CIs are 95% bootstrap ($n_{\text{boot}} = 5000$, seed = 20260603).

Classification used three ordered groups (low/medium/high) defined by strict score thresholds at the 33rd/67th percentile boundaries (scores strictly above the boundary assigned to the higher group). Because the motor composite is a discrete 0–10 sum, participants tied at the boundary score are assigned to the lower group, so the realised high group is typically smaller than one third of the sample (22–32% depending on dataset); actual proportions are reported alongside each κ estimate. Weighted Cohen's κ (linear weights) was used. Both the full-scale total and the overlap-free rest score are reported as criteria; the latter quantifies classification agreement not attributable to shared items. Binary cutoff sweeps (8%–33%) provide AUC, sensitivity, specificity, PPV, and NPV. All-pairs analysis computed r_{full} and r_{rest} for all $\binom{10}{2} = 45$ two-item combinations in the PCS/SWASH-style 10-item samples. External validity tested motor pair vs. full scale against five criteria with no item overlap with the PCS.

3 Results

3.1 Convergence

The Norms sample provides the cleanest context comparison: participants were randomly assigned to PCS or SWASH from a common pool, so the correlation difference is attributable to the administered context—including hypnotic induction and associated framing—rather than cohort differences. Motor composite correlations were higher under SWASH ($r_{\text{full}} = 0.849$, $r_{\text{rest}} = 0.694$) than PCS ($r_{\text{full}} = 0.783$, $r_{\text{rest}} = 0.565$); Fisher's r-to-z confirms the difference (r_{full}: $z = -2.18$, $p = .029$; r_{rest}: $z = -2.36$, $p = .018$). Hypnotic induction amplifies the motor-pair coupling; both contexts produce substantial correlations.

Table 2 and Figure 1 show the full pattern. Across three PCS datasets, the motor composite correlated $r_{\text{full}} = 0.78$–0.79 ($R^2 = 0.61$–0.62) and $r_{\text{rest}} = 0.55$–0.58, replicating tightly across independent cohorts ($N = 244$–508). SWASH shows higher coupling: $r_{\text{full}} = 0.81$–0.85, $r_{\text{rest}} = 0.63$–0.69. Cross-dataset PCS–SWASH comparisons are non-significant ($p = .24$, $.38$), consistent with between-cohort variability dominating in separate samples. HGSHS:A involuntariness falls between ($r_{\text{full}} = 0.80$, $r_{\text{rest}} = 0.66$). HGSHS:A objective binary is lower ($r_{\text{full}} = 0.68$), consistent with binary scoring discarding within-item variation.

Table 2: Motor composite correlations with full-scale total and rest score.

Sample	N	r_{full}	R^2	95% CI	r_{rest}	95% CI	SB
Norms-PCS	244	0.783	0.61	[0.732, 0.827]	0.565	[0.474, 0.644]	0.61
PCS-VVIQ	508	0.778	0.61	[0.741, 0.811]	0.549	[0.483, 0.612]	0.63
vEAR-PCS	452	0.785	0.62	[0.748, 0.819]	0.579	[0.514, 0.641]	0.62
Norms-SWASH	240	0.849	0.72	[0.814, 0.880]	0.694	[0.624, 0.754]	0.79
SWASH-val	418	0.807	0.65	[0.774, 0.838]	0.630	[0.570, 0.687]	0.75
HGSHS:A (inv.)	584	0.799	0.64	[0.768, 0.827]	0.659	[0.610, 0.705]	0.70
HGSHS:A (obj.)	584	0.682	0.46	[0.639, 0.720]	0.449	[0.384, 0.511]	0.42
HGSHS-5:G (prior short form)	—	*0.83*	*0.69*	—	—	—	—

Note. r_{full}: motor items inside total. r_{rest}: motor pair vs. non-motor rest score. SB: Spearman-Brown reliability. CIs: 95% bootstrap ($n_{\text{boot}} = 5000$). All $p < .001$. HGSHS-5:G: Riegel et al. (2021); Zech et al. (2024).

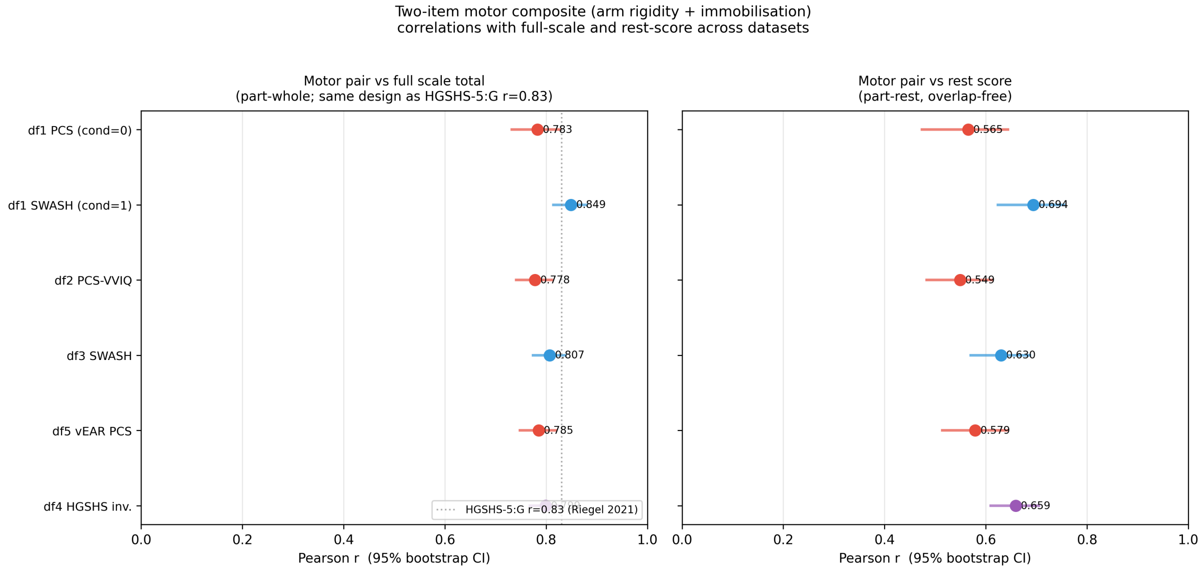

Figure 1: Forest plot: motor composite r_{full} (left) and r_{rest} (right) with 95% CIs across datasets. Red = PCS, blue = SWASH, purple = HGSHS:A. Dotted line: HGSHS-5:G $r = 0.83$.

3.2 Floor effects

Figure 2 shows per-item means and floor rates. Music hallucination and negative visual hallucination are near floor in every dataset: 80–93% at zero, means 0.14–0.43/5. The motor items sit near mid-scale (means 2.0–3.2/5, floor rates 5–12%). With near-universal zero scores, these items add little to between-person variance on the total, which is therefore disproportionately shaped by the items that retain between-person spread. This is consistent with bifactor analysis of HGSHS:A (Zahedi and Sommer, 2022).

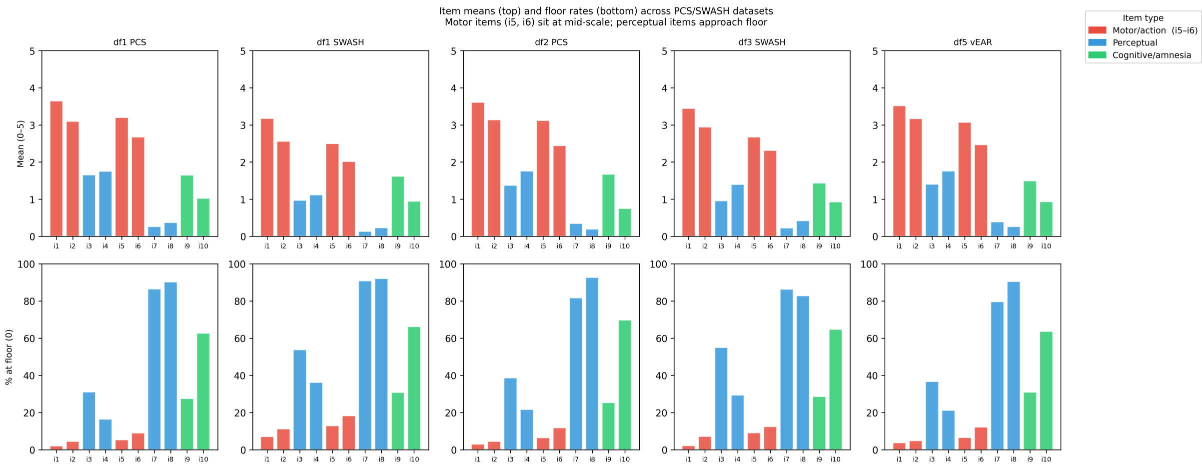

Figure 2: Per-item means (top) and floor rates (bottom) across five intensity datasets. Red = motor items (i5, i6); blue = perceptual; green = cognitive/amnesia.

3.3 External validity

Table 3 and Figure 3 show correlations with external criteria sharing no items with the PCS. The motor pair retains 71–80% of the full scale's correlation with vEAR and anomalous experiences, the two criteria where both scales show nonzero signal. On both, the non-motor rest score predicts outcomes slightly more strongly than the motor pair (vEAR: $r_{\text{rest}} = 0.354$ vs. $r_{\text{motor}} = 0.297$; anomalous: $r_{\text{rest}} = 0.189$ vs. $r_{\text{motor}} = 0.155$); the rest score carries eight items to the pair's two. The motor pair indexes the shared first component; the perceptual and cognitive items carry domain-specific variance beyond it. Flow and DES correlations are near-zero for both the motor pair and the full scale; retention percentages for those criteria are not meaningful.

Table 3: External validity: motor pair vs. full scale (no item overlap).

Criterion	N	r_{motor}	r_{rest}	r_{full}	Retained
vEAR (video-rated)	452	0.297	0.354	0.372	80%
Anomalous experiences	497	0.155	0.189	0.202	77%
Past-life experiences	497	0.158	0.214	0.221	71%
Flow (combined)	432	0.109	0.131	0.139	78%†
DES dissociation	559	0.100	0.092	0.107	—†

† Both values near zero; retention percentage uninformative.

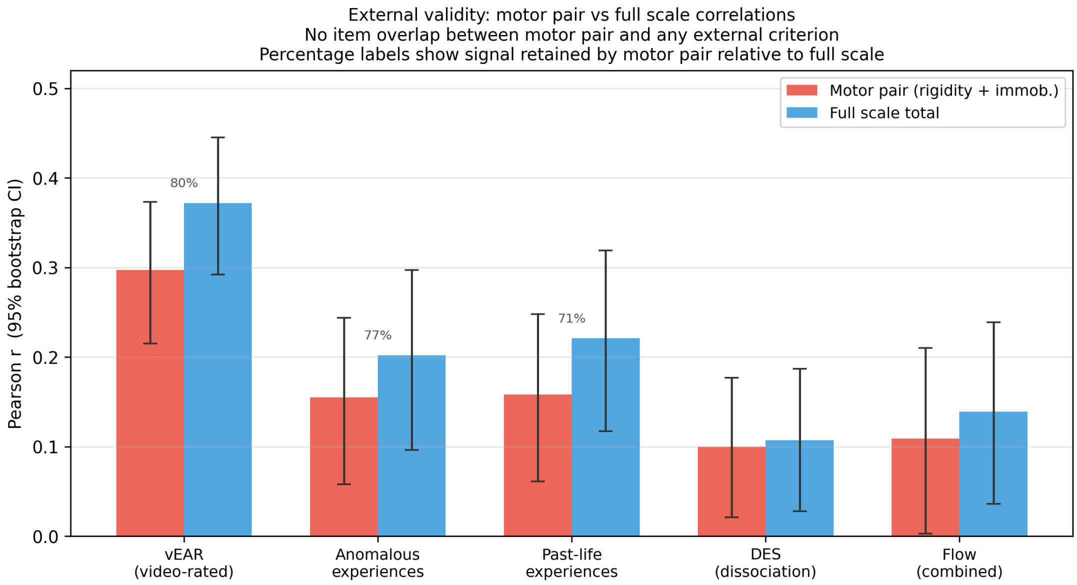

Figure 3: Motor pair (red) vs. full scale (blue) with external criteria. Error bars: 95% bootstrap CIs. Labels: % of full-scale r retained.

3.4 Classification

Table 4 shows three-group weighted κ and AUC for high-group classification (strict threshold at the 67th percentile; realised proportions 22–32% depending on dataset). Against the full total, PCS reaches $\kappa_w = 0.50$–0.55 and SWASH/HGSHS:A involuntariness reaches $\kappa_w = 0.65$. The overlap-free rest-score criterion produces lower values—PCS 0.29–0.34, SWASH 0.45–0.46, HGSHS:A involuntariness 0.52—quantifying the portion of agreement attributable to shared items. AUC ranges from 0.83–0.86 (PCS) to 0.89–0.91 (SWASH/HGSHS:A involuntariness). Binary cutoff sweeps show high specificity (0.87–0.99) across the full 8–33% range; sensitivity is modest at tight cutoffs.

Table 4: Classification: weighted κ and AUC.

		Full total		Rest score		
Sample	N	κ_w	κ_u	κ_w	κ_u	AUC
Norms-PCS	244	0.548	0.446	0.325	0.262	0.829
PCS-VVIQ	508	0.504	0.400	0.294	0.203	0.852
vEAR-PCS	452	0.522	0.410	0.335	0.240	0.858
Norms-SWASH	240	0.657	0.550	0.459	0.355	0.905
SWASH-val	418	0.652	0.547	0.448	0.363	0.896
HGSHS:A (inv.)	584	0.651	0.563	0.522	0.442	0.893
HGSHS:A (obj.)	584	0.554	0.459	0.347	0.267	0.841
HGSHS-5:G (prior short form)	—	*≈0.58*				

Note. AUC: high-group on full total (threshold at 67th percentile; realised proportion varies by dataset). Rest score: non-motor items. HGSHS-5:G: Riegel et al. (2021); Zech et al. (2024).

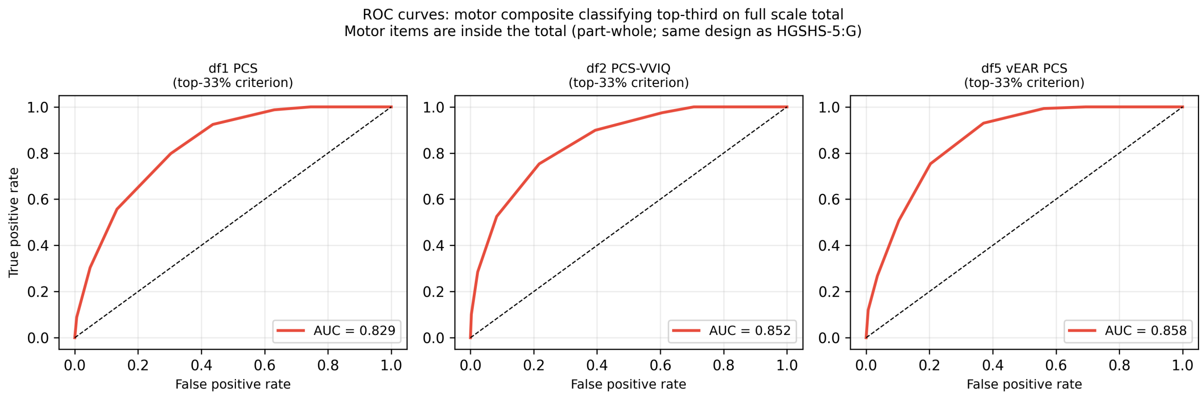

Figure 4: ROC curves: motor composite classifying high-group on full total (threshold at 67th percentile), three PCS datasets. AUC = 0.829, 0.852, 0.858.

3.5 All-pairs analysis

Figure 5 shows where the motor pair falls among all 45 possible two-item combinations per dataset. By r_{full}, the motor pair ranks first or second in all three PCS datasets and fourth in the Norms-SWASH dataset; in the SWASH validation dataset it ranks 13th, where the top pair is taste + arm rigidity ($r_{\text{full}} = 0.845$). The r_{rest} ranking is weaker—the motor pair drops to ranks 2, 9, 11, 8, and 23 across the five datasets. A motor item appears in the top pair in every dataset; perceptual floor items never do.

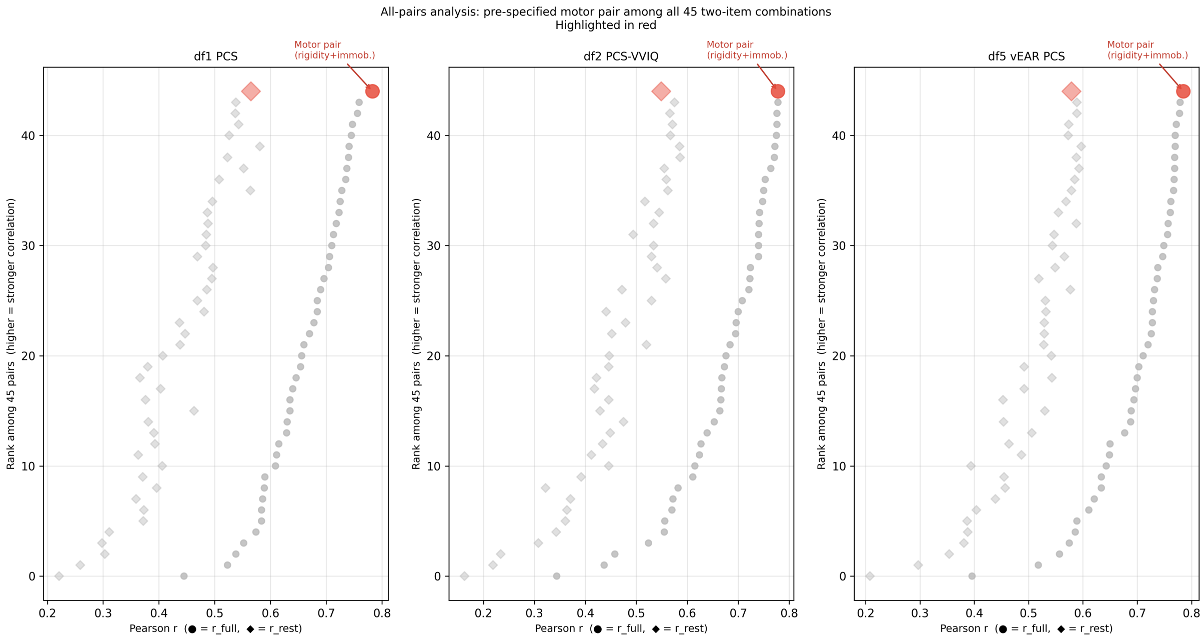

Figure 5: All 45 two-item combinations ranked by r_{full} (circles) and r_{rest} (diamonds) across three PCS datasets. Motor pair in red.

3.6 Temporal stability

Embedded motor-pair test-retest: $r = 0.338$ ($n = 61$, PCS; full scale: $r = 0.523$; SWASH: $r = 0.497$). Overlap-free motor-T1 to rest-T2: $r = 0.222$ (PCS, ns). Both figures are from embedded administrations within the full ten-item procedure; standalone two-item retest stability is unknown and may differ in either direction.

4 Discussion

Among all 45 two-item subsets of the established ten-item pool, the motor pair is the top-ranked or near-top-ranked pair in every dataset, with a motor item in the leading pair regardless of scale or context; the optimality claim is bounded to this comparison class. The structure—variance concentrating in motor-challenge items—holds across non-hypnotic (PCS), hypnotic (SWASH), and classic group (HGSHS:A) administration and across intensity and involuntariness axes. The magnitude of that concentration differs by context: the randomised Norms comparison, where the same pool was assigned to PCS or SWASH, establishes that stronger motor-pair coupling under induction reflects the administered context rather than cohort composition (r_{full}: $p = .029$; r_{rest}: $p = .018$).

Floor effects explain the high r_{full} values. In unselected samples, music hallucination and negative visual hallucination score zero for the great majority of participants; with near-zero variance they contribute little to between-person rank-ordering on the total, which is therefore dominated by the motor items. The near-floor perceptual items are difficult suggestions carrying domain-specific variance that the two-item screen gives up; the rest score, with eight items to the pair's two, also carries a reliability advantage that accounts for part of its slightly higher external signal.

High specificity (0.87–0.99 across the binary sweep) means that high motor-pair scores have a low false-positive rate for high-group status; modest sensitivity at tight cutoffs means some true high responders will not be flagged. That trade-off is appropriate for first-pass triage, not final selection.

Four limits bound the interpretation. These are embedded administrations: standalone two-item performance is untested. Floor concentration is sample-dependent: in enriched high-responding samples the proxy advantage will shrink. HGSHS:A indexes involuntariness and objective pass/fail, not subjective intensity; its results are not directly comparable with the PCS and SWASH findings. All analyses are secondary reanalysis of open data; no new data were collected.

Open Science Statement

Data. All datasets are on OSF: Norms sample: `osf.io/et85n`; PCS-VVIQ sample: `osf.io/aeyhw`; SWASH validation sample: `osf.io/wujk8`; HGSHS:A sample: `osf.io/jfmc4`; vEAR-PCS sample: `osf.io/a2skp`; external criteria: `osf.io/u5wjy`.

Code. `analysis.ipynb`, `lib_data.py`, and analysis scripts `a01`–`a08` at `https://github.com/Wordweaver-EH/Hypnoscales`.

Author contributions (CRediT). [To be completed at submission.]

Supplementary Materials

S1. Analysis code

All analysis code is available at `https://github.com/Wordweaver-EH/Hypnoscales`. The pipeline consists of the following files:

- `lib_data.py` — Shared data-loading, canonical scoring, exclusion logic, motor-pair column mappings, and bootstrap CI functions. All analysis scripts import from this file; nothing is hard-coded elsewhere.
- `a01_descriptives.py` — Per-item means, SDs, floor and ceiling rates, and dataset summary table (`table_item_descriptives.csv`, `table_dataset_summary.csv`).
- `a02_convergence.py` — Motor composite correlations (r_{full}, r_{rest}, Spearman-Brown reliability) with bootstrap CIs and Fisher r-to-z context comparisons (`table_main_convergence.c`
- `a03_retest.py` — Test-retest correlations for the motor pair, full scale, and overlap-free motor-T1 to rest-T2 (`table_test_retest.csv`).
- `a04_classification.py` — Weighted Cohen's κ (three threshold groups), AUC, and binary cutoff sweep for sensitivity, specificity, PPV, and NPV (`table_classification.csv`).
- `a05_all_pairs.py` — r_{full} and r_{rest} for all $\binom{10}{2} = 45$ two-item combinations in the PCS/SWASH-style 10-item samples (`table_all_pairs.csv`).
- `a06_external.py` — Motor pair and full-scale correlations with five external criteria (no item overlap) (`table_external_validity.csv`).
- `a07_sensitivity.py` — Sensitivity checks: itemwise z-scored composites, listwise deletion, alternative cutoffs (`table_sensitivity.csv`).
- `a08_figures.py` — All publication figures (Figures 1–5).
- `analysis.ipynb` — Self-contained Jupyter notebook that reproduces the full pipeline inline, with inline narrative.

S2. Source datasets

- **Norms sample** (PCS + SWASH, between-subjects): `osf.io/et85n`. Lush et al. (2021).
- **PCS-VVIQ sample**: `osf.io/aeyhw`. Lush et al. (2021).
- **SWASH validation sample**: `osf.io/wujk8`. Lush et al. (2018).
- **HGSHS:A sample**: `osf.io/jfmc4`. Acunzo and Terhune (2021).
- **vEAR-PCS sample** (external validity, convergence): `osf.io/a2skp`.
- **External criteria** (anomalous experiences, flow, DES, past-life, vEAR behavioural ratings): `osf.io/u5wjy`.

References

Acunzo, D. J. and Terhune, D. B. (2021). Measuring hypnotic suggestibility beyond the Stanford scales: a psychometric analysis. *Psychological Assessment*, 33(9):816–829.

Bowers, K. S. (1981). Do the Stanford scales tap the "classic suggestion effect"? *International Journal of Clinical and Experimental Hypnosis*, 29(1):42–53.

Coe, W. C. and Sarbin, T. R. (1971). An alternative interpretation to the multiple composition of hypnotic scales: a rejoinder to Hilgard. *Psychological Bulletin*, 76(3):200–209.

Eisinga, R., te Grotenhuis, M., and Pelzer, B. (2013). The reliability of a two-item scale: Pearson, Cronbach, or Spearman-Brown? *International Journal of Public Health*, 58(4):637–642.

Lush, P., Moga, G., McLatchie, N., and Dienes, Z. (2018). The sussex-waterloo scale of hypnotizability (SWASH): measuring capacity for altering conscious experience. *Neuroscience of Consciousness*, 2018(1):niy006.

Lush, P., Scott, R. B., Seth, A. K., and Dienes, Z. (2021). The phenomenological control scale: Measuring the capacity for creating illusory nonvolition, hallucination and delusion. *Collabra: Psychology*, 7(1):29542.

Riegel, M., Pyza, M., Wierzbicki, A., Szekeres, M., Wicher, B., Wróbel, A., and Holas, P. (2021). HGSHS-5:G – a short form of the Harvard Group Scale of Hypnotic Susceptibility. *Frontiers in Psychology*, 12:622468.

Sadler, P. and Woody, E. Z. (2004). Is hypnotic suggestibility a stable trait? Conclusions drawn from a large sample and a brief measure. *International Journal of Clinical and Experimental Hypnosis*, 52(4):395–418.

Shor, R. E. and Orne, E. C. (1962). *Harvard Group Scale of Hypnotic Susceptibility, Form A*. Consulting Psychologists Press, Palo Alto, CA.

Zahedi, A. and Sommer, W. (2022). Bifactor structure of the Harvard Group Scale of Hypnotic Susceptibility. *International Journal of Clinical and Experimental Hypnosis*, 70(2):137–157.

Zech, M. et al. (2024). HGSHS-5:G—first results with the short version of the test for the Harvard Group Scale of Hypnotic Susceptibility and a comparison with the full version. *Frontiers in Psychology*, 15:1422920.